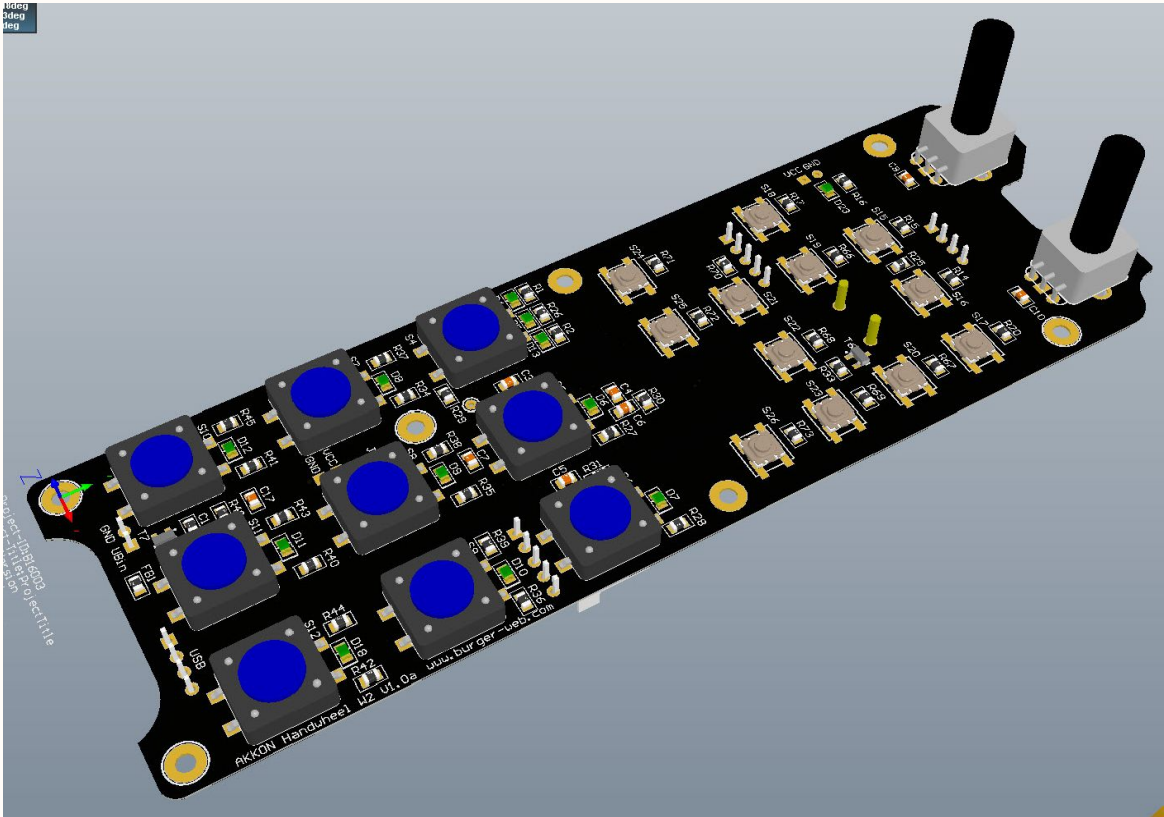


Project:AKKON_HandWheel_W3_V1_1a.PrjPcb
Project number: B16003



Journal		
Version	Date	Changes
V1.0	25.01.2017	first version
V1.1	03.06.2017	Changed: J7 CAN connector Changed: voltage divider for detection of emergency switches Chanted: added serial resistor R13 to MCLR for protection of T1 if RTS = 1 no firmware changes to Version 1.0

Title: Overview	
Project: AKKON HandWheel_W3_V1_1a.PrjPcb	
Author: Gerhard Burger	
Approved:*	
Date: 03.06.2017	Time: 12:53:59
Page: 1 of 8	
Version: 1.0	
File: Overview.SchDoc	
Project ID: B16003	
BURGER www.burger-web.com	

A

A

B

B

C

C

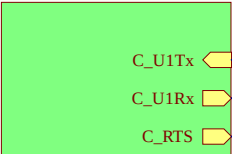
D

D

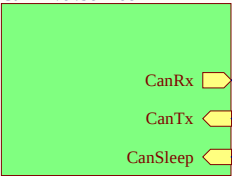
Controller
Controller.SchDoc

User interface
UserInterface.SchDoc

USB Driver
USBDriver.SchDoc



CAN Driver
CanDriver.SchDoc



Power
Power.SchDoc



Project overview
Overview.SchDoc

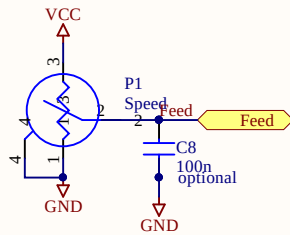


PCB1
AKKON Handwheel W3 V1.1

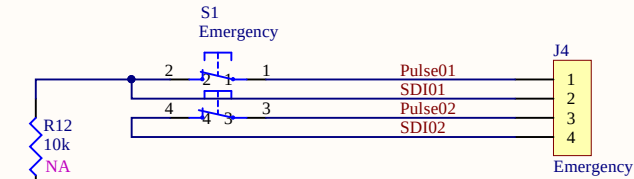
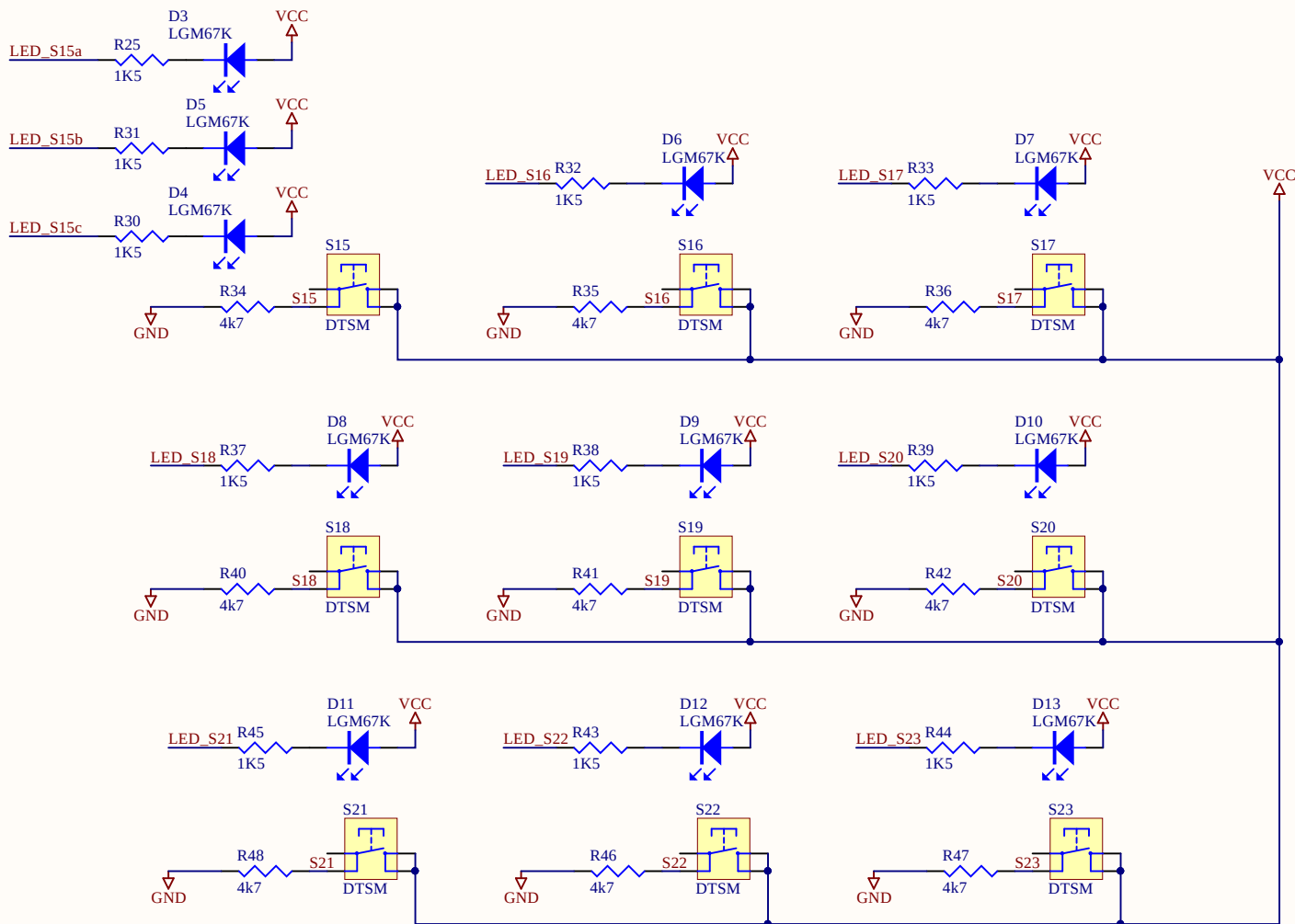
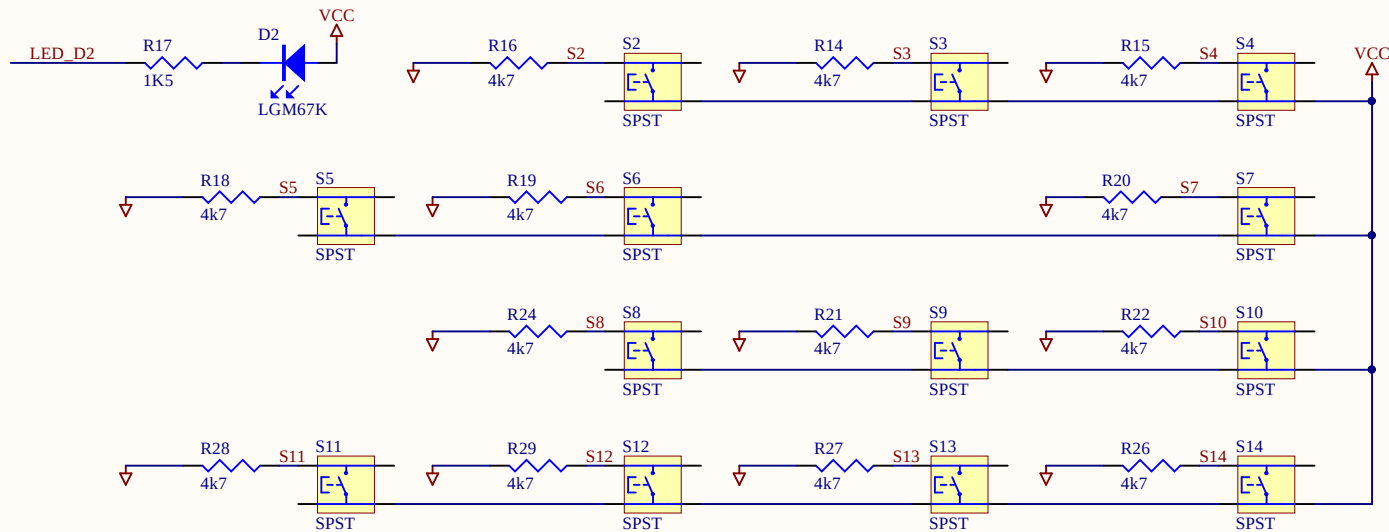
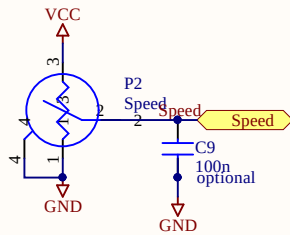


Place nylon washers on mounting holes between case and PCB
E.g. washer M2.5, Duratol, nylon, Farnell 7047472

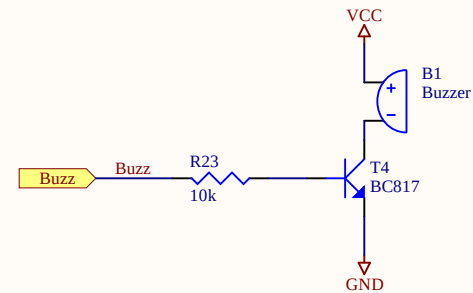
Title: Architecture		Project ID: B16003
Project: AKKON HandWheel W3 V1_1a.PrjPch		
Author: Gerhard Burger		
Approved:*		
Date: 03.06.2017	Time: 12:53:59	
Page: 2 of 8		 www.burger-web.com
Version: 1.1		
File: Handwheel.SchDoc		



Alternative: 53UAD-T22-B15L



Voltage divider at $U_{in} = 24VDC$
 $U_{aEmerg} = 4.89 V$
 $I_{aEmerg} = 0.4 mA$
Alternative: $24VDC \rightarrow 5V$ 100 uA 24k / 96K



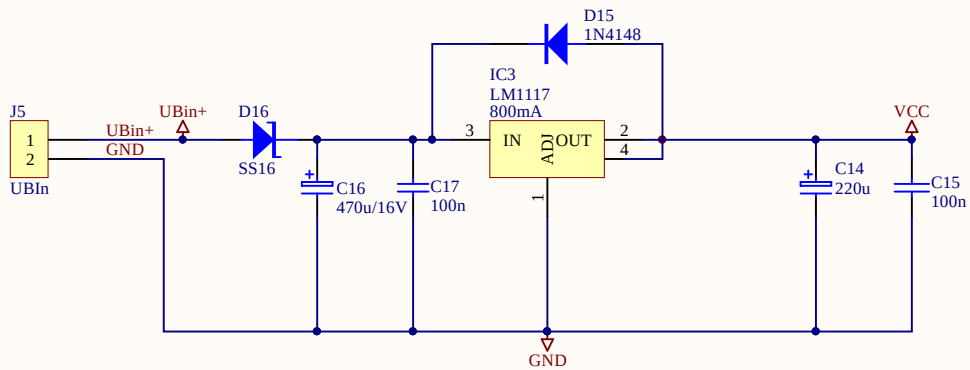
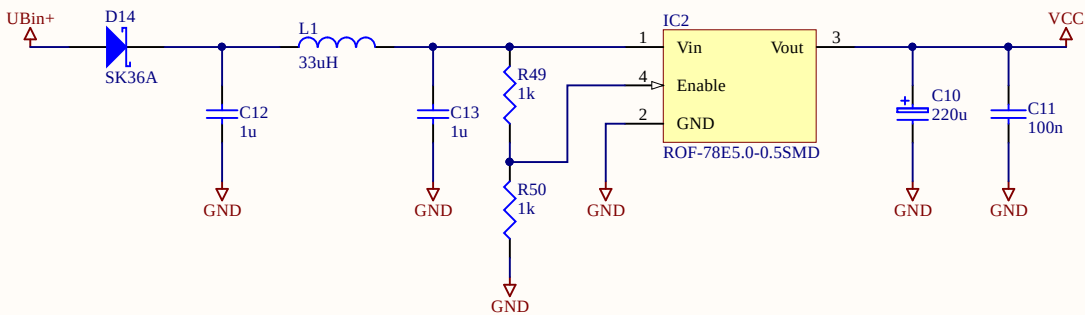
S13	S13
S15	S15
S16	S16
S17	S17
S18	S18
S19	S19
S20	S20
S21	S21
S22	S22

S7	S7
S4	S4
S3	S3
S2	S2
S6	S6
S8	S8
S9	S9
S10	S10
S5	S5
S11	S11
S14	S14
S12	S12

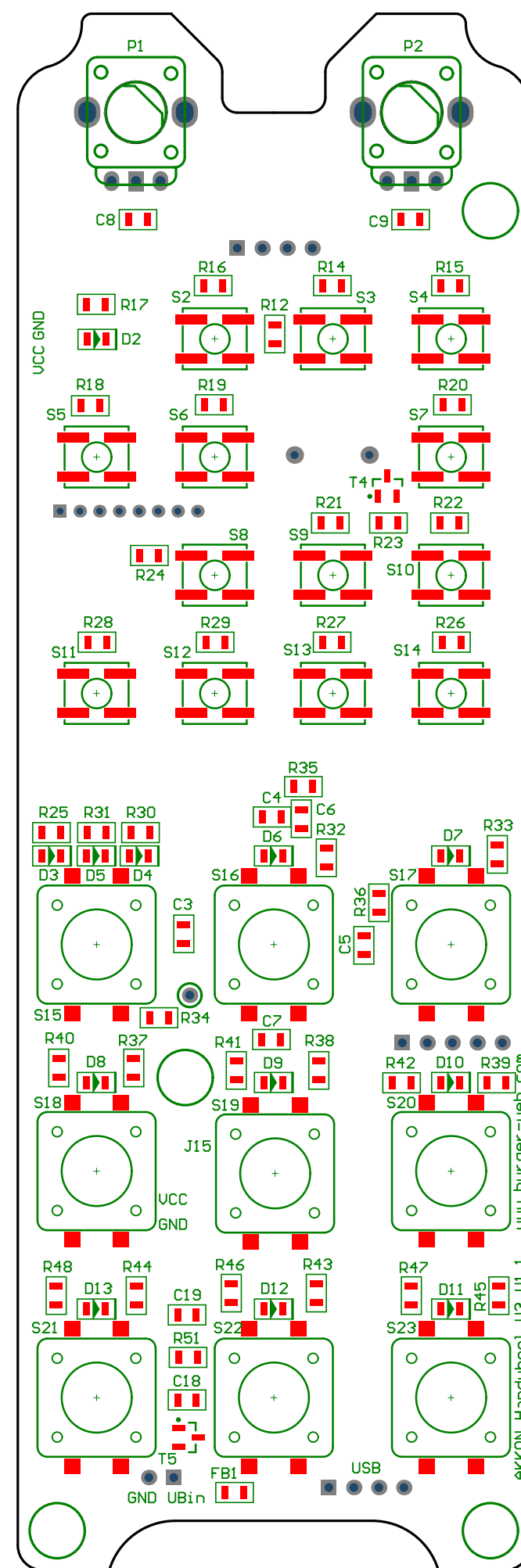
LED_D2	LED_D2
LED_S15a	LED_S15a
LED_S15b	LED_S15b
LED_S15c	LED_S15c
LED_S16	LED_S16
LED_S17	LED_S17
LED_S18	LED_S18
LED_S19	LED_S19
LED_S20	LED_S20
LED_S21	LED_S21
LED_S22	LED_S22

Title: User interface	
Project: AKKON HandWheel_W3_V1_1a.Prj	
Author: Gerhard Burger	
Approved:*	
Date: 03.06.2017	Time: 12:53:59
Page: 4 of 8	
Version: 1.0	
File: UserInterface.SchDoc	

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Title: Power			 www.burger-web.com
Project: AKKON HandWheel W3_V1_1a.Prj			
Author: Gerhard Burger			
Approved:*			
Date: 03.06.2017	Time: 12:53:59		
Page: 6 of 8			
Version: 1.0			
File: Power.SchDoc			



Project-ID:B16003
Project-Title:AKKON Handwheel W3
Version:1.1
Date:03.06.2017
Size: 52 mm * 158 mm
Material: FR4, 1.6 mm



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ENGINEER:

PCB DESIGNER:

DATE:
03.06.2017

FILE NAME:
Handwheel.PcbDoc

TITLE:

AKKON Handwheel

PART NO.:
B16003

DWG NO: XXXXXXXXXXXX

REV:

SCALE:

A

A

B

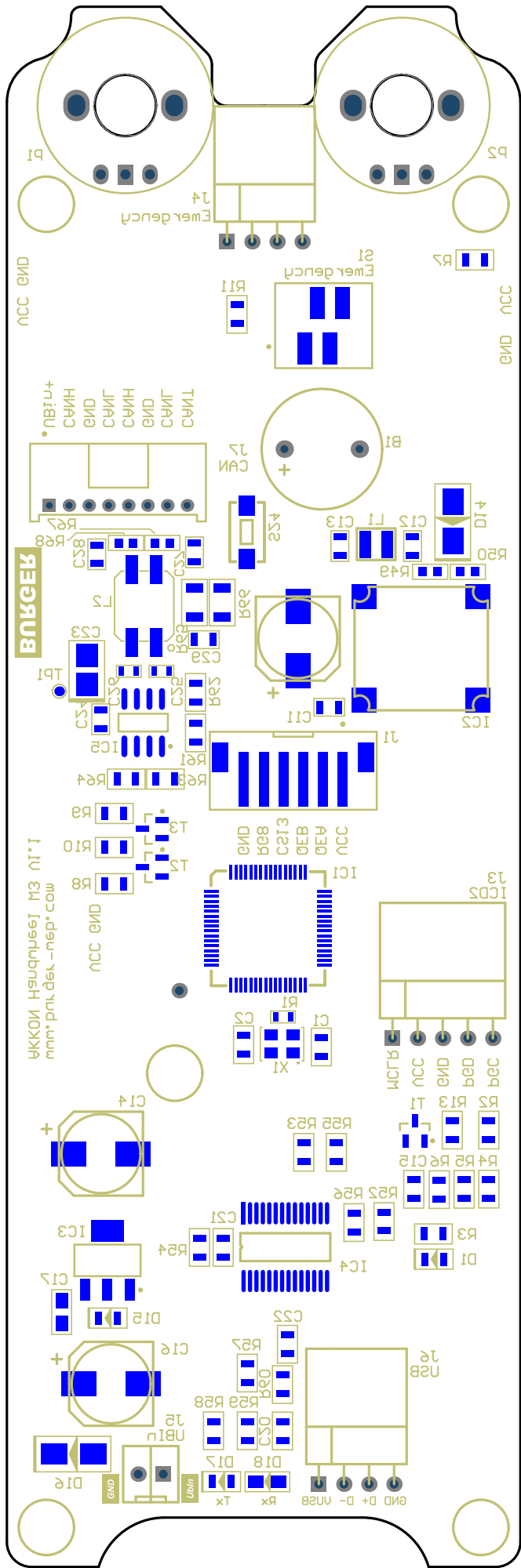
B

C

C

D

D



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ENGINEER:	TITLE: AKKON Handwheel		
PCB DESIGNER:			
DATE: 03.06.2017	PART NO.: B16003	REV:	
FILE NAME: Handwheel.PcbDoc	DWG NO: XXXXXXXXXXXX	SCALE:	

